F18CF – Linear Algebra

Tutorial Sheet 6: Rank, Nullity, and Isomorphisms

Getting started.

1. Starting with the kernel, we solve by Gaussian elimination:

$$\left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 2 & 1 & -1 & 2 & 0 \\ 2 & -1 & 3 & -1 & 0 \\ 1 & 1 & -2 & 4 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 3 & -5 & 0 & 0 \\ 0 & -2 & 4 & -3 & 0 \\ 0 & 2 & -4 & 3 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & -1 & 2 & 1 & 0 \\ 0 & 3 & -5 & 0 & 0 \\ 0 & 0 & 2 & -9 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The only free variable is x_4, so we set $x_4 = \alpha$, and then we get by substitution $x_3 = 9\alpha/2$, $x_2 = 15\alpha/2$, $x_1 = -5\alpha/2$. The general solution is of the form $\frac{1}{2}(-5, 15, 9, 2)^T$, so a basis for the kernel is

$$\begin{pmatrix} -5 \\ 15 \\ 9 \\ 2 \end{pmatrix}.$$

Moreover, since the first three columns of the row echelon form correspond to the pivot variables, it follows that a basis of the image is given by the first three columns of the original matrix, that is,

$$\begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 3 \\ -2 \end{pmatrix}.$$

Therefore the rank is 3 and the nullity is 1.

For the next matrix, we apply the same procedure:

$$\left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 0 \\ 2 & -1 & 3 & 3 & 0 \\ -1 & 3 & -4 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 0 \\ 0 & -5 & 5 & -5 & 0 \\ 0 & 5 & -5 & 5 & 0 \\ 0 & 1 & -1 & 1 & 0 \end{array}\right) \to \left(\begin{array}{cccc|c} 1 & 2 & -1 & 4 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

We get two free variables, x_3, x_4, so we set $x_3 = \alpha$, $x_4 = \beta$, and by substitution we get $x_2 = \alpha - \beta$ and $x_1 = -2(\alpha - \beta) + \alpha - 4\beta = -\alpha - 2\beta$. The general solution is thus of the form

$$\begin{pmatrix} -\alpha - 2\beta \\ \alpha - \beta \\ \alpha \\ \beta \end{pmatrix} = \alpha \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

so a basis for the kernel would be

$$\begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

Moreover, since the first two columns of the row echelon form correspond to the pivot variables, the first two columns of the original matrix form a basis of the range space:

$$\begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -1 \\ 3 \\ 1 \end{pmatrix}$$

Therefore the rank is 2 and the nullity is 2.

2. Let A be a 3×4 matrix with rk(A) = 3.

   (a) By the Rank-Nullity Theorem, the dimension of $\ker(A)$ is $\dim(\mathbb{R}^4) - \text{rk(A)} = 1$.

   (b) The range has dimension $3 = \dim(\mathbb{R}^3)$ so A is surjective by a result of the course, i.e. $\text{im}(A) = \mathbb{R}^3$. In particular, $A\vec{x} = \vec{b}$ is consistent for all $\vec{b} \in \mathbb{R}^3$.

   (c) $A\vec{x} = \vec{b}$ would have infinitely many solutions. There are two ways of seeing this. The first one is to notice that, when computing a row echelon form of the matrix, we would find 3 pivot variables, 1 free variable, and no forbidden row (since there are as many pivots as the number of rows). Another way to see this is through the observation that you can always add a vector from the kernel to a solution of the equation $A\vec{x} = \vec{b}$ to get another solution of that same equation. Since the kernel contains infinitely many vectors, the equation $A\vec{x} = \vec{b}$ would have infinitely many solutions.

   Let A be a 4×3 matrix with rk(A) = 3.

   (a) By the Rank-Nullity Theorem, the dimension of $\ker(A)$ is $\dim(\mathbb{R}^3) - \text{rk(A)} = 0$.

   (b) The range again has dimension $3 < \dim(\mathbb{R}^4)$, so $\text{im}(A)$ is a strict (3-dimensional) subspace of $\mathbb{R}^4$. Therefore $A\vec{x} = \vec{b}$ is not consistent for all $\vec{b} \in \mathbb{R}^4$ (it is only consistent when $\vec{b}$ lies in the image of A).

(c) $A\vec{x} = \vec{b}$ would have either no solution or a unique solution. This is because there are no vectors in the null space apart from the zero vector.

To see this more thoroughly, you can suppose $\vec{x}_1$ and $\vec{x}_2$ are solutions. Then $A(\vec{x}_1 - \vec{x}_2) = \vec{b} - \vec{b} = \underline{0}$. But since the kernel has dimension 0, $\vec{x}_1 - \vec{x}_2 = \underline{0}$ hence $\vec{x}_1 = \vec{x}_2$, and the solution is unique.

3. (a) The matrix for the transformation

$$T : \mathbb{R}^3 \to \mathbb{R}^3, \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} x+y+z \\ x+2y+4z \\ x+4y+16z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

in the basis $(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix})$ is simply $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix}$

(b) If our basis is now $(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix})$, then the matrix of change of basis is

$$P = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

We compute its inverse using the standard algorithm:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & -3 & -1 & 1 & -2 & 0 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & -4 & 1 & 1 & -3 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & -4 & 1 & 1 & -3 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & -1 & 2 \\ 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1/4 & -1/4 & 3/4 \end{array}\right)$$

$$\to \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & -1/4 & -1/4 \\ 0 & 1 & 0 & -1/4 & 3/4 & -1/4 \\ 0 & 0 & 1 & -1/4 & -1/4 & 3/4 \end{array}\right),$$

so the inverse of P is

$$P^{-1} = \begin{pmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{pmatrix}.$$

The matrix of T in the new basis is thus

$$B = P^{-1}AP = \begin{pmatrix} 3/4 & -1/4 & -1/4 \\ -1/4 & 3/4 & -1/4 \\ -1/4 & -1/4 & 3/4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} -9/2 & -11/2 & -9 \\ -1/2 & -1/2 & -2 \\ 27/2 & 31/2 & 24 \end{pmatrix}$$

4. **An image problem.**

$$T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R}), \quad P(x) \mapsto P(x+1) - P(x).$$

(a) Let $P(x)$ and $Q(x)$ belong to $\mathcal{P}_n(\mathbb{R})$ and $\lambda \in \mathbb{R}$.

$$T(P(x) + Q(x)) = P(x+1) - P(x) + Q(x+1) - Q(x) = T(P(x)) + T(Q(x))$$

$$T(\lambda P(x)) = \lambda P(x+1) - \lambda P(x) = \lambda T(P(x))).$$

Hence T is a linear map.

(b) Let $P(x) = a_0 + a_1 x - \cdots +_n x^n$. We have

$$T(P(x)) = a_0 + a_1(x+1) + \cdots + a_n(x+1)^n - a_0 - a_1 x - \cdots - a_n x^n.$$

First notice that $T(P(x))$ is of degree at most $n-1$ since the only terms contributing to degree n are $a_n(x+1)^n$ and $-a_n x^n$ and we see using the Binomial Theorem that the terms in x^n cancel out.

Next, notice that the leading term of $T(P(x))$ is exactly $na_n x^{n-1}$. Indeed, $a_0 + a_1(x+1) + \cdots + a_{n-1}(x+1)^{n-1} - a_0 - a_1 x - \cdots - a_{n-1}x^{n-1}$ is of degree at most $n-2$ by the same reasoning applied to $a_0 + a_1(x+1) + \cdots + a_{n-1}(x+1)^{n-1}$, and what is left of $T(P(x))$ is

$$a_n(x+1)^n - a_n x^n = na_n x^{n-1} + \underbrace{(\ldots)}_{\text{degree} \leq n-2},$$

as can be seen using the Binomial Theorem. Thus, for a polynomial $P(x)$ of degree $n \geq 1$, we have deg $T(P(x)) = n-1$. In particular, $T(P(x))$ is not the zero polynomial.

Thus, if $T(P(x)) = 0$ for every x, then deg $P = 0$, hence P is constant. The converse is immediate. Thus, $\ker(T)$ is the subspace of constant polynomials, which has dimension 1.

(c) By the Rank-Nullity theorem, the dimension of $\mathrm{im}(T)$ is $\dim(\mathcal{P}_n(\mathbb{R})) - \dim(\ker(T)) = (n+1) - 1 = n$.

(d) For $P \in \mathcal{P}_n(\mathbb{R})$, $T(P(x))$ has no x^n term as explained above, therefore $R(T) \subset \mathcal{P}_{n-1}(\mathbb{R})$. Since from the previous question we have $\dim R(T) = n = \dim \mathcal{P}_{n-1}(\mathbb{R})$, it follows that $R(T) = \mathcal{P}_{n-1}(\mathbb{R})$ (two nested subspaces of the same dimension are equal).

5. **Magic squares.** A 3×3 *homogeneous magic square* is a 3×3 matrix such that the sum of the coefficients in each row and each column is 0.

(a) Clearly the zero matrix 0_3 is in W. Let $A, B \in W$ and $\lambda \in \mathbb{R}$. Since $A, B \in W$, we have for every $1 \leq i_0, j_0 \leq 3$:

$$\sum_{1 \leq i \leq 3} a_{i,j_0} = 0, \quad \sum_{1 \leq i \leq 3} b_{i,j_0} = 0, \quad \sum_{1 \leq i \leq 3} a_{i,j_0} = 0, \quad \sum_{1 \leq i \leq 3} b_{i,j_0} = 0$$

Consider $A+B$, whose (i,j)-entry is $a_{i,j}+b_{i,j}$. Then for $1 \leq i_0, j_0 \leq 3$, we have for every $1 \leq i_0, j_0 \leq 3$:

$$\sum_{1 \leq i \leq 3} a_{i,j_0} + b_{i,j_0} = 0 + 0 = 0$$

and

$$\sum_{1 \leq j \leq 3} a_{i_0,j} + b_{i_0,j} = 0 + 0 = 0$$

so $A+B \in W$.

Similarly, consider λA, whose (i,j)-entry is $\lambda a_{i,j}$. We have for every $1 \leq i_0, j_0 \leq 3$:

$$\sum_{1 \leq i \leq 3} \lambda a_{i,j_0} = \lambda \times 0 = 0, \quad \sum_{1 \leq j \leq 3} \lambda a_{i_0,j} = \lambda \times 0 = 0$$

so $\lambda A \in W$.

Therefore W is a subspace of $M_3(\mathbb{R})$.

(b)

$$T : W \to \mathbb{R}^4, A \mapsto (a_{11}, a_{12}, a_{21}, a_{22})^T.$$

T is a linear map since

$$T(A+B) = (a_{11}+b_{11}, a_{12}+b_{12}, a_{21}+b_{21}, a_{22}+b_{22})^T = T(A) + T(B)$$

and

$$T(\lambda A) = (\lambda a_{11}, \lambda a_{12}, \lambda a_{21}, \lambda a_{22})^T = \lambda T(A).$$

So T is linear.

(c) Let us show that T is bijective. Let $\vec{a} := (a_{11}, a_{12}, a_{21}, a_{22})^T \in \mathbb{R}^4$. To get a magic square A such that $T(A) = \vec{a}$, the conditions on the first two rows and first two columns imply that A is of the form

$$A := \begin{pmatrix} a_{11} & a_{12} & -a_{11}-a_{12} \\ a_{21} & a_{22} & -a_{21}-a_{22} \\ -a_{11}-a_{21} & -a_{12}-a_{22} & * \end{pmatrix}.$$

Moreover, to be a magic square, the condition on the third row and third column completely determine the last coefficient, and we get:

$$A := \begin{pmatrix} a_{11} & a_{12} & -a_{11}-a_{12} \\ a_{21} & a_{22} & -a_{21}-a_{22} \\ -a_{11}-a_{21} & -a_{12}-a_{22} & a_{11}+a_{12}+a_{21}+a_{22} \end{pmatrix}.$$

Conversely, the above matrix is indeed a homogeneous magic square, Thus, there is exactly one solution to the equation $T(X) = \vec{a}$, so T is bijective.

Therefore T is an isomorphism, and we have $\dim(W) = \dim(\mathbb{R}^4) = 4$.

6. **Sums of consecutive powers.**

(a) Using the formula $P_n(x) := \frac{x(x-1)\cdots(x-n+1)}{n!}$
$\mathcal{B}' = (1, x, \frac{x(x-1)}{2}, \frac{x(x-1)(x-2)}{6})$ forms a basis of $\mathcal{P}_3(\mathbb{R})$ by a theorem of the course, as the degree of P_i is i for every $0 \leq i \leq 3$.

(b) We have

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}x^2 - \frac{1}{2}x, \quad P_3(x) = \frac{1}{6}x^3 - \frac{1}{2}x^2 + \frac{1}{3}x.$$

Thus, we get

$$[\mathcal{B}']_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1/2 & 1/3 \\ 0 & 0 & 1/2 & -1/2 \\ 0 & 0 & 0 & 1/6 \end{pmatrix}.$$

To compute $[\mathcal{B}]_{\mathcal{B}'}$, we invert this matrix using the standard algorithm:

$$\left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1/2 & 1/3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1/2 & -1/2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1/6 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1/2 & 1/3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 6 \end{array}\right)$$

$$\to \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1/2 & 0 & 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 6 \end{array}\right) \to \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 6 \end{array}\right).$$

Therefore, $[\mathcal{B}]_{\mathcal{B}'} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \end{pmatrix}$

(c) First, we treat the cases $i = 0$ and $i = 1$. We have

$$P_1(x+1) - P_1(x) = (x+1) - x = 1 = P_0(x),$$

and

$$P_2(x+1) - P_2(x) = \frac{(x+1)x}{2} - \frac{x(x-1)}{2} = x = P_1(x).$$

For $i \geq 2$, we get, starting from the left hand side

$$P_{i+1}(x+1) - P_{i+1}(x) = \frac{(x+1)x(x-1)\cdots(x-i+1)}{(i+1)!} - \frac{x(x-1)\cdots(x-i)}{(i+1)!}$$

$$= \frac{x(x-1)\cdots(x-i+1)}{(i+1)!}(x+1-(x-i)) = \frac{x(x-1)\cdots(x-i+1)}{i!} = P_i(x).$$

Thus, for $n \geq 1$ and $i \geq 0$ we have (cancelling the terms in the sum)

$$\sum_{k=0}^{n} P_i(k) = \sum_{k=0}^{n} P_{i+1}(k+1) - P_{i+1}(k)$$
$$= P_{i+1}(n+1) - P_{i+1}(0) = P_{i+1}(n+1).$$

(d) To compute the sum, we replace k, k^2, k^3 by their expression in the basis $P_0, \dots, P_3$, using the matrix $[\mathcal{B}]_{\mathcal{B}'}$. We have

$$x = P_1(x), \quad x^2 = 2P_2(x) + P_1(x), \quad x^3 = 6P_3(x) + 6P_2(x) + P_1(x).$$

We thus get, using the formula from the previous question:
We have

$$\sum_{k=1}^{n} k = \sum_{k=1}^{n} P_1(k) = P_2(n+1) = \frac{n(n+1)}{2}.$$

We have

$$\sum_{k=1}^{n} k^2 = \sum_{k=1}^{n} (2P_2(k)+P_1(k)) = 2P_3(n+1)+P_2(n+1) = 2\frac{(n+1)n(n-1)}{6}+\frac{n(n+1)}{2},$$

thus,

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

We have

$$\sum_{k=1}^{n} k^3 = \sum_{k=1}^{n} (6P_3(k) + 6P_2(k) + P_1(k)) = 6P_4(n+1) + 6P_3(n+1) + P_2(n+1)$$

$$= 6\frac{(n+1)n(n-1)(n-2)}{24} + 6\frac{(n+1)n(n-1)}{6} + \frac{n(n+1)}{2}$$
$$= \frac{(n+1)n\left((n-1)(n-2) + 4(n-1) + 2\right)}{4},$$

that is,

$$\sum_{k=1}^{n} k^3 = \frac{(n+1)^2 n^2}{4}.$$